

## Practical: LVM Configuration (Comprehensive)

*Can use any available disk (like /dev/sda, /dev/sdb) instead of /dev/nvme0n1, depending on your system's device naming.*

### Question

You are given a disk /dev/sdb with free space. Perform the following tasks:

#### Physical Volume (PV)

1. Create two partitions on /dev/sdb:
  - /dev/sdb1 → 4GB
  - /dev/sdb2 → 4GB
2. Initialise both partitions as Physical Volumes
3. Verify PV creation

#### Volume Group (VG)

1. Create a Volume Group named vgdata using both PVs
2. Verify VG details
3. Display:
  - Total size
  - Free space
  - Physical Extent (PE) size (default)
4. Create another Volume Group vgcustom with custom PE size (e.g., 16MB)

#### Logical Volumes (LV)

1. Create a Logical Volume lvsize of 2GB in vgdata (size-based)
2. Create a Logical Volume lvext using Logical Extents (e.g., percentage of free space)
3. Create a Logical Volume lvcustom in vgcustom using extents

#### Formatting (Different File Systems)

- Format:
  - lvsize → XFS
  - lvext → xfs
  - lvnext → xfs

#### Mounting

##### Temporary:

- Mount:
  - lvsize → /mnt/data1

- lvnext → /mnt/data2

### **Permanent:**

- Mount lvnext → /mnt/backup using UUID in /etc/fstab

### **Verification**

- Verify:
  - Physical Volumes
  - Volume Groups
  - Logical Volumes
  - Filesystem types
  - Mounted filesystems

### **LVM Management (Resize Operations)**

1. Extend lvsize by 1GB
2. Resize the filesystem accordingly
3. Reduce lvext by 500MB (after unmounting)
4. Resize the filesystem safely before reduction

### **Cleanup (WITHOUT reboot)**

1. Unmount all logical volumes
2. Remove entries from /etc/fstab
3. Remove Logical Volumes
4. Remove Volume Groups
5. Remove Physical Volumes
6. Delete partitions
7. Apply changes

-----command-----

# =====

# STEP 1: Create partitions (4GB + 4GB + extra for custom VG)

# =====

**fdisk /dev/sdb**

# n → p → 1 → Enter → +4G

# n → p → 2 → Enter → +4G

# n → p → 3 → Enter → +2G

# t → 1 → 8e

# t → 2 → 8e

# t → 3 → 8e

# w

partprobe

lsblk

# =====

# STEP 2: Create Physical Volumes

# =====

pvcreate /dev/sdb1 /dev/sdb2 /dev/sdb3

# Verify

pvs

pvdisplay

# =====

# STEP 3: Create Volume Groups

# =====

vgcreate vgdata /dev/sdb1 /dev/sdb2

vgcreate -s 16M vgcustom /dev/sdb3

# Verify

vgs

vgdisplay

# =====

# STEP 4: Create Logical Volumes

# =====

lvcreate -L 2G -n lvsize vgdata

lvcreate -l 50%FREE -n lvext vgdata

```
lvcreate -l 100%FREE -n lvnext vgcustom
```

```
# Verify
```

```
lvs
```

```
lvdisplay
```

```
# =====
```

```
# STEP 5: Format Filesystems (all xfs as per your work)
```

```
# =====
```

```
mkfs.xfs /dev/vgdata/lvsize
```

```
mkfs.xfs /dev/vgdata/lvext
```

```
mkfs.xfs /dev/vgcustom/lvnext
```

```
# =====
```

```
# STEP 6: Mount (Temporary)
```

```
# =====
```

```
mkdir -p /mnt/data1 /mnt/data2 /mnt/backup
```

```
mount /dev/vgdata/lvsize /mnt/data1
```

```
mount /dev/vgcustom/lvnext /mnt/data2
```

```
# =====
```

```
# STEP 7: Permanent Mount
```

```
# =====
```

```
blkid
```

```
vi /etc/fstab
```

```
# Add:
```

```
# UUID=xxxx /mnt/backup xfs defaults 0 0
```

```
systemctl daemon-reload
```

```
mount -a
```

```
# =====  
# STEP 8: Verification  
# =====  
  
pvs  
vgs  
lvs  
lsblk -f  
df -h  
  
# =====  
# STEP 9: Extend LV  
# =====  
  
lvextend -L +1G /dev/vgdata/lvsize  
xfs_growfs /mnt/data1  
  
# =====  
# STEP 10: Reduce LV (correct safe method)  
# =====  
  
umount /dev/vgdata/lvext  
lvreduce -L 1.5G /dev/vgdata/lvext  
mkfs.xfs -f /dev/vgdata/lvext  
mount /dev/vgdata/lvext /mnt/data2  
  
# =====  
# STEP 11: Cleanup (WITHOUT reboot)  
# =====  
  
umount /mnt/data1  
umount /mnt/data2  
umount /mnt/backup  
  
vi /etc/fstab  
# remove added entry
```

```
lvremove /dev/vgdata/lvsize -y
lvremove /dev/vgdata/lvext -y
lvremove /dev/vgcustom/lvnext -y
```

```
vgrename vgdata
vgrename vgcustom
```

```
pvremove /dev/sdb1 /dev/sdb2 /dev/sdb3
```

```
fdisk /dev/sdb
# d → delete all → w
```

```
partprobe
```

```
lsblk
```

-----X-----

```
04:41:35 | root@localhost | ~ | cmd:149 | cpu: 3.51 | mem:23%
🚀 lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                  8:0    0   25G  0 disk
├─sda1               8:1    0    1M  0 part
├─sda2               8:2    0    2G  0 part /boot
└─sda3               8:3    0   23G  0 part
    ├─cs-root        253:0    0 20.5G  0 lvm  /
    └─cs-swap        253:1    0  2.5G  0 lvm  [SWAP]
sdb                  8:16    0   10G  0 disk
sr0                 11:0    1 1024M  0 rom
```

## Physical Volume (PV)

4. Create two partitions on /dev/sdb:

- /dev/sdb1 → 4GB
- /dev/sdb2 → 4GB

```
04:41:39 | root@localhost | ~ | cmd:150 | cpu: 3.47 | mem:25% | disk:21% | ip: | git: | s
🚀 fdisk /dev/sdb
```

```
Welcome to fdisk (util-linux 2.40.2).
```

```
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.
```

```
Command (m for help): p
```

```
Disk /dev/sdb: 10 GiB, 10737418240 bytes, 20971520 sectors
```

```
Disk model: VMware Virtual S
```

```
Units: sectors of 1 * 512 = 512 bytes
```

```
Sector size (logical/physical): 512 bytes / 512 bytes
```

```
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disklabel type: dos
```

```
Disk identifier: 0x17c80010
```

```
Command (m for help): n
```

```
Partition type
```

```
  p   primary (0 primary, 0 extended, 4 free)
```

```
  e   extended (container for logical partitions)
```

```
Select (default p): p
```

```
Partition number (1-4, default 1): 1
```

```
First sector (2048-20971519, default 2048):
```

```
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-20971519, default 20971519): +4G
```

```
Created a new partition 1 of type 'Linux' and of size 4 GiB.
```

```
Partition #1 contains a xfs signature.
```

```
Do you want to remove the signature? [Y]es/[N]o: yes
```

```
The signature will be removed by a write command.
```

```
Command (m for help): t
Selected partition 1
Hex code or alias (type L to list all): 8e
Changed type of partition 'Linux' to 'Linux LVM'.
```

```
Command (m for help): p
Disk /dev/sdb: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x17c80010
```

| Device    | Boot | Start | End     | Sectors | Size | Id | Type      |
|-----------|------|-------|---------|---------|------|----|-----------|
| /dev/sdb1 |      | 2048  | 8390655 | 8388608 | 4G   | 8e | Linux LVM |

```
Filesystem/RAID signature on partition 1 will be wiped.
```

```
Command (m for help): n
Partition type
  p   primary (1 primary, 0 extended, 3 free)
  e   extended (container for logical partitions)
Select (default p): p
Partition number (2-4, default 2): 2
First sector (8390656-20971519, default 8390656):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (8390656-20971519, default 20971519): +4G
```

```
Created a new partition 2 of type 'Linux' and of size 4 GiB.
```

```
Command (m for help): p
Disk /dev/sdb: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x17c80010
```

| Device    | Boot | Start   | End      | Sectors | Size | Id | Type      |
|-----------|------|---------|----------|---------|------|----|-----------|
| /dev/sdb1 |      | 2048    | 8390655  | 8388608 | 4G   | 8e | Linux LVM |
| /dev/sdb2 |      | 8390656 | 16779263 | 8388608 | 4G   | 83 | Linux     |

```
Filesystem/RAID signature on partition 1 will be wiped.
```

Command (m for help): t  
Partition number (1,2, default 2): 2  
Hex code or alias (type L to list all): l

|                    |                    |                    |                    |
|--------------------|--------------------|--------------------|--------------------|
| 00 Empty           | 27 Hidden NTFS Win | 82 Linux swap / So | c1 DRDOS/sec (FAT- |
| 01 FAT12           | 39 Plan 9          | 83 Linux           | c4 DRDOS/sec (FAT- |
| 02 XENIX root      | 3c PartitionMagic  | 84 OS/2 hidden or  | c6 DRDOS/sec (FAT- |
| 03 XENIX usr       | 40 Venix 80286     | 85 Linux extended  | c7 Syrix           |
| 04 FAT16 <32M      | 41 PPC PReP Boot   | 86 NTFS volume set | da Non-FS data     |
| 05 Extended        | 42 SFS             | 87 NTFS volume set | db CP/M / CTOS / . |
| 06 FAT16           | 4d QNX4.x          | 88 Linux plaintext | de Dell Utility    |
| 07 HPFS/NTFS/exFAT | 4e QNX4.x 2nd part | 8e Linux LVM       | df BootIt          |
| 08 AIX             | 4f QNX4.x 3rd part | 93 Amoeba          | e1 DOS access      |
| 09 AIX bootable    | 50 OnTrack DM      | 94 Amoeba BBT      | e3 DOS R/O         |
| 0a OS/2 Boot Manag | 51 OnTrack DM6 Aux | 9f BSD/OS          | e4 SpeedStor       |
| 0b W95 FAT32       | 52 CP/M            | a0 IBM Thinkpad hi | ea Linux extended  |
| 0c W95 FAT32 (LBA) | 53 OnTrack DM6 Aux | a5 FreeBSD         | eb BeOS fs         |
| 0e W95 FAT16 (LBA) | 54 OnTrackDM6      | a6 OpenBSD         | ee GPT             |
| 0f W95 Ext'd (LBA) | 55 EZ-Drive        | a7 NeXTSTEP        | ef EFI (FAT-12/16/ |
| 10 OPUS            | 56 Golden Bow      | a8 Darwin UFS      | f0 Linux/PA-RISC b |
| 11 Hidden FAT12    | 5c Priam Edisk     | a9 NetBSD          | f1 SpeedStor       |
| 12 Compaq diagnost | 61 SpeedStor       | ab Darwin boot     | f4 SpeedStor       |
| 14 Hidden FAT16 <3 | 63 GNU HURD or Sys | af HFS / HFS+      | f2 DOS secondary   |
| 16 Hidden FAT16    | 64 Novell Netware  | b7 BSDI fs         | f8 EBBR protective |
| 17 Hidden HPFS/NTF | 65 Novell Netware  | b8 BSDI swap       | fb VMware VMFS     |
| 18 AST SmartSleep  | 70 DiskSecure Mult | bb Boot Wizard hid | fc VMware VMKCORE  |
| 1b Hidden W95 FAT3 | 75 PC/IX           | bc Acronis FAT32 L | fd Linux raid auto |
| 1c Hidden W95 FAT3 | 80 Old Minix       | be Solaris boot    | fe LANstep         |
| 1e Hidden W95 FAT1 | 81 Minix / old Lin | bf Solaris         | ff BBT             |
| 24 NEC DOS         |                    |                    |                    |

Aliases:

|          |      |
|----------|------|
| linux    | - 83 |
| swap     | - 82 |
| extended | - 05 |
| uefi     | - EF |
| raid     | - FD |
| lvm      | - 8E |
| linuxex  | - 85 |

Hex code or alias (type L to list all): 8E

Changed type of partition 'Linux' to 'Linux LVM'.

```
Command (m for help): p
Disk /dev/sdb: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x17c80010
```

| Device    | Boot | Start   | End      | Sectors | Size | Id | Type      |
|-----------|------|---------|----------|---------|------|----|-----------|
| /dev/sdb1 |      | 2048    | 8390655  | 8388608 | 4G   | 8e | Linux LVM |
| /dev/sdb2 |      | 8390656 | 16779263 | 8388608 | 4G   | 8e | Linux LVM |

Filesystem/RAID signature on partition 1 will be wiped.

```
Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
```

```
04:47:07 | root@localhost | ~ | cmd:151 | cpu: 0.52 | mem:25% | disk:21% | ip: | git: | ssh:
🚀 partprobe
```

```
04:47:14 | root@localhost | ~ | cmd:152 | cpu: 0.47 | mem:26% | disk:21% | ip: | git: | ssh:
🚀 lsblk
```

| NAME        | MAJ:MIN | RM | SIZE  | RO | TYPE | MOUNTPOINTS |
|-------------|---------|----|-------|----|------|-------------|
| sda         | 8:0     | 0  | 25G   | 0  | disk |             |
| ├─sda1      | 8:1     | 0  | 1M    | 0  | part |             |
| ├─sda2      | 8:2     | 0  | 2G    | 0  | part | /boot       |
| ├─sda3      | 8:3     | 0  | 23G   | 0  | part |             |
| │ └─cs-root | 253:0   | 0  | 20.5G | 0  | lvm  | /           |
| │ └─cs-swap | 253:1   | 0  | 2.5G  | 0  | lvm  | [SWAP]      |
| sdb         | 8:16    | 0  | 10G   | 0  | disk |             |
| ├─sdb1      | 8:17    | 0  | 4G    | 0  | part |             |
| └─sdb2      | 8:18    | 0  | 4G    | 0  | part |             |
| sr0         | 11:0    | 1  | 1024M | 0  | rom  |             |

## 5. Initialise both partitions as Physical Volumes

```
04:49:13 | root@localhost | ~ | cmd:153 | cpu: 0.20 | mem:26
git: | ssh: | time:0s
🚀 pvcreate /dev/sdb1
Physical volume "/dev/sdb1" successfully created.

04:56:00 | root@localhost | ~ | cmd:154 | cpu: 0.11 | mem:26
git: | ssh: | time:0s
🚀 pvcreate /dev/sdb2
Physical volume "/dev/sdb2" successfully created.
```

## 6. Verify PV creation

```
04:56:05 | root@localhost | ~ | cmd:155 | cpu: 0.26 | mem:27% | disk:2
git: | ssh: | time:0s
🚀 pvs
PV          VG Fmt  Attr PSize  PFree
/dev/sda3   cs  lvm2 a--  <23.00g  0
/dev/sdb1    lvm2 ---   4.00g  4.00g
/dev/sdb2    lvm2 ---   4.00g  4.00g

04:56:45 | root@localhost | ~ | cmd:156 | cpu: 0.13 | mem:27% | disk:2
git: | ssh: | time:0s
🚀 pvdisplay
--- Physical volume ---
PV Name           /dev/sda3
VG Name           cs
PV Size           <23.00 GiB / not usable 0
Allocatable       yes (but full)
PE Size           4.00 MiB
Total PE          5887
Free PE           0
Allocated PE      5887
PV UUID           JvsA7G-0Iq1-x020-iLGb-yedU-K6uc-VWRD31

"/dev/sdb1" is a new physical volume of "4.00 GiB"
--- NEW Physical volume ---
PV Name           /dev/sdb1
VG Name
PV Size           4.00 GiB
```

## Volume Group (VG)

5. Create a Volume Group named vgdata using both PVs

```
05:01:57 | root@localhost | ~ | cmd:162 | cpu: 0.13 | mem:27% | disk:21% |  
git: | ssh: | time:0s  
🚀 vgcreate vgdata /dev/sdb1 /dev/sdb2  
Volume group "vgdata" successfully created
```

6. Verify VG details

```
05:02:14 | root@localhost | ~ | cmd:163 | cpu: 0.10 | mem:27% | disk:21% |  
git: | ssh: | time:0s  
🚀 vgs  
VG      #PV #LV #SN Attr   VSize   VFree  
cs       1  2  0 wz--n- <23.00g    0  
vgdata   2  0  0 wz--n-  7.99g 7.99g
```

7. Display:

- Total size
- Free space
- Physical Extent (PE) size (default)

```
git: | ssh: | time:0s  
🚀 vgsdisplay  
--- Volume group ---  
VG Name                cs  
System ID  
Format                 lvm2  
Metadata Areas         1  
Metadata Sequence No   3  
VG Access              read/write  
VG Status              resizable  
MAX LV                 0  
Cur LV                2  
Open LV                2  
Max PV                 0  
Cur PV                1  
Act PV                1  
VG Size                <23.00 GiB  
PE Size                4.00 MiB  
Total PE               5887  
Alloc PE / Size        5887 / <23.00 GiB  
Free PE / Size         0 / 0  
VG UUID                72HaA1-sMQc-Aoe2-xpAT-utp  
  
--- Volume group ---  
VG Name                vgdata  
System ID  
Format                 lvm2  
Metadata Areas         2
```

8. Create another Volume Group vgcustom with custom PE size (e.g., 16MB)

```
05:03:49 | root@localhost | ~ | cmd:165 | cpu: 0.02 | mem:
🚀 fdisk /dev/sdb
```

Command (m for help): p

**Disk /dev/sdb: 10 GiB, 10737418240 bytes, 20971520 sectors**

Disk model: VMware Virtual S

Units: sectors of 1 \* 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes

Disklabel type: dos

Disk identifier: 0x17c80010

| Device    | Boot | Start    | End      | Sectors | Size | Id | Type      |
|-----------|------|----------|----------|---------|------|----|-----------|
| /dev/sdb1 |      | 2048     | 8390655  | 8388608 | 4G   | 8e | Linux LVM |
| /dev/sdb2 |      | 8390656  | 16779263 | 8388608 | 4G   | 8e | Linux LVM |
| /dev/sdb3 |      | 16779264 | 20971519 | 4192256 | 2G   | 83 | Linux     |

Command (m for help): t

Partition number (1-3, default 3):

Hex code or alias (type L to list all): 8e

Changed type of partition 'Linux' to 'Linux LVM'.

Command (m for help): p

**Disk /dev/sdb: 10 GiB, 10737418240 bytes, 20971520 sectors**

Disk model: VMware Virtual S

Units: sectors of 1 \* 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes

Disklabel type: dos

Disk identifier: 0x17c80010

| Device    | Boot | Start    | End      | Sectors | Size | Id | Type      |
|-----------|------|----------|----------|---------|------|----|-----------|
| /dev/sdb1 |      | 2048     | 8390655  | 8388608 | 4G   | 8e | Linux LVM |
| /dev/sdb2 |      | 8390656  | 16779263 | 8388608 | 4G   | 8e | Linux LVM |
| /dev/sdb3 |      | 16779264 | 20971519 | 4192256 | 2G   | 8e | Linux LVM |

Command (m for help): w

The partition table has been altered.

Calling ioctl() to re-read partition table.

Syncing disks.

```
05:06:39 | root@localhost | ~ | cmd:166 | cpu: 0.58 | mem:27%
🚀 partprobe
```

```
05:13:09 | root@localhost | ~ | cmd:170 | cpu: 0.27 | mem:27% | disk:21% | ip: | git: | ssh: | time:0s
🚀 vgcreate -s 16M vgcustom /dev/sdb3

Volume group "vgcustom" successfully created
```

```
05:13:27 | root@localhost | ~ | cmd:172 | cpu: 0.32 | mem:27% | disk:21% | ip: | git: | ssh: | time:0s
🚀 vgs
```

| VG       | #PV | #LV | #SN | Attr   | VSize   | VFree |
|----------|-----|-----|-----|--------|---------|-------|
| cs       | 1   | 2   | 0   | wz--n- | <23.00g | 0     |
| vgcustom | 1   | 0   | 0   | wz--n- | 1.98g   | 1.98g |
| vgdata   | 2   | 0   | 0   | wz--n- | 7.99g   | 7.99g |

## Logical Volumes (LV)

4. Create a Logical Volume lvsize of 2GB in vgdata (size-based)

```
05:13:35 | root@localhost | ~ | cmd:173 | cpu: 0.29 | mem:27% | disk:21% | ip: | git: | ssh: | time:0s
🚀 lvcreate -L 2G -n lvsize vgdata
Logical volume "lvsize" created.
```

5. Create a Logical Volume lvext using Logical Extents (e.g., percentage of free space)
6. Create a Logical Volume lvcustom in vgcustom using extents

```
05:15:02 | root@localhost | ~ | cmd:174 | cpu: 0.24 | mem:27% | disk:21% | ip: | git: | ssh: | time:0s
🚀 lvcreate -l 50%FREE -n lvnext vgdata
Logical volume "lvnext" created.
```

```
05:19:02 | root@localhost | ~ | cmd:175 | cpu: 0.07 | mem:28% | disk:21% | ip: | git: | ssh: | time:0s
🚀 lvcreate -l 50 -n lvnext vgcustom
Logical volume "lvnext" created.
```

```
05:19:31 | root@localhost | ~ | cmd:176 | cpu: 0.41 | mem:27% | disk:21% | ip: | git: | ssh: | time:0s
🚀 lvs
```

| LV                   | VG       | Attr       | LSize   | Pool | Origin | Data% | Meta% |
|----------------------|----------|------------|---------|------|--------|-------|-------|
| Log Cpy%Sync Convert |          |            |         |      |        |       |       |
| root                 | cs       | -wi-ao---- | <20.50g |      |        |       |       |
| swap                 | cs       | -wi-ao---- | 2.50g   |      |        |       |       |
| lvnext               | vgcustom | -wi-a----- | 800.00m |      |        |       |       |
| lvnext               | vgdata   | -wi-a----- | <3.00g  |      |        |       |       |
| lvsize               | vgdata   | -wi-a----- | 2.00g   |      |        |       |       |

## Formatting (Different File Systems)

- Format:
  - lvsize → XFS
  - lvext → xfs
  - lvcustom → xfs

```
05:21:04 | root@localhost | ~ | cmd:179 | cpu: 0.24 | mem:28% | disk:21% | ip: | git: | ssh: | time:0s
🔥 mkfs.xfs /dev/vgdata/lvsize
meta-data=/dev/vgdata/lvsize      isize=512    agcount=4, agsize=131072 blks
=                               sectsz=512   attr=2, projid32bit=1
=                               crc=1       finobt=1, sparse=1, rmapbt=1
=                               reflink=1   bigtime=1 inobtcount=1 nnext64=1
=                               exchange=0  metadir=0
data      =                       bsize=4096  blocks=524288, imaxpct=25
=                               sunit=0        swidth=0 blks
naming    =version 2             bsize=4096  ascii-ci=0, ftype=1, parent=0
log       =internal log         bsize=4096  blocks=16384, version=2
=                               sectsz=512    sunit=0 blks, lazy-count=1
realtime  =none                 extsz=4096  blocks=0, rtextents=0
=                               rgcount=0     rgspace=0 extents
=                               zoned=0        start=0 reserved=0

05:23:48 | root@localhost | ~ | cmd:180 | cpu: 0.22 | mem:28% | disk:21% | ip: | git: | ssh: | time:0s
🔥 mkfs.xfs /dev/vgdata/lvnext
meta-data=/dev/vgdata/lvnext      isize=512    agcount=4, agsize=196352 blks
=                               sectsz=512   attr=2, projid32bit=1
=                               crc=1       finobt=1, sparse=1, rmapbt=1
=                               reflink=1   bigtime=1 inobtcount=1 nnext64=1
=                               exchange=0  metadir=0
data      =                       bsize=4096  blocks=785408, imaxpct=25
=                               sunit=0        swidth=0 blks
naming    =version 2             bsize=4096  ascii-ci=0, ftype=1, parent=0
log       =internal log         bsize=4096  blocks=16384, version=2
=                               sectsz=512    sunit=0 blks, lazy-count=1
realtime  =none                 extsz=4096  blocks=0, rtextents=0
=                               rgcount=0     rgspace=0 extents
=                               zoned=0        start=0 reserved=0

05:23:59 | root@localhost | ~ | cmd:181 | cpu: 0.19 | mem:28% | disk:21% | ip: | git: | ssh: | time:0s
🔥 mkfs.xfs /dev/vgcustom/lvnext
meta-data=/dev/vgcustom/lvnext    isize=512    agcount=4, agsize=51200 blks
=                               sectsz=512   attr=2, projid32bit=1
=                               crc=1       finobt=1, sparse=1, rmapbt=1
=                               reflink=1   bigtime=1 inobtcount=1 nnext64=1
=                               exchange=0  metadir=0
```

## Mounting

### Temporary:

- Mount:
  - lvsize → /mnt/data1
  - lvext → /mnt/data2

```
05:24:21 | root@localhost | ~ | cmd:182 | cpu: 0.65 | mem:28% | disk:21%  
🚀 mkdir -p /mnt/{data1,data2,backup}  
  
05:25:19 | root@localhost | ~ | cmd:183 | cpu: 0.42 | mem:28% | disk:21%  
🚀 mount /dev/vgdata/lvsize /mnt/data1  
  
05:26:05 | root@localhost | ~ | cmd:184 | cpu: 0.20 | mem:28% | disk:21%  
🚀 mount /dev/vgdata/lvnext /mnt/data2
```

### Permanent:

- Mount lvnext → /mnt/backup using UUID in /etc/fstab

```
#  
# /etc/fstab  
# Created by anaconda on Sun Mar 15 19:41:01 2026  
#  
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.  
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.  
#  
# After editing this file, run 'systemctl daemon-reload' to update systemd  
# units generated from this file.  
#  
UUID=e79aa976-2f04-4e51-8942-08170334e649 / xfs defaults 0 0  
UUID=b41c8b19-dfe8-4088-8806-ec8fa03b629a /boot xfs defaults 0 0  
UUID=de55cc71-342c-4646-817d-ae6a99e96439 none swap defaults 0 0  
  
#/dev/sda1 /mnt/sda1 xfs defaults 0 0  
#-----  
  
/dev/vgcustom/lvnext /mnt/backup xfs defaults 0 0  
~  
~  
~  
~  
~  
~  
~  
~  
~  
~
```

```
05:34:50 | root@localhost | ~ | cmd:195 | cpu: 0.37 | mem:28% | disk:21% | ip:  
🚀 systemctl daemon-reload  
  
05:35:42 | root@localhost | ~ | cmd:196 | cpu: 0.33 | mem:28% | disk:21% | ip:  
🚀 mount -a
```

## Verification

- Verify:
  - Physical Volumes
  - Volume Groups
  - Logical Volumes
  - Filesystem types
  - Mounted filesystems

```
05:35:47 | root@localhost | ~ | cmd:197 | cpu: 0.43 | mem:28% | disk:21% | ip: | git: | ssh: | time:0s
🚀 vgs
VG          #PV #LV #SN Attr   VSize  VFree
cs          1  2  0 wz--n- <23.00g  0
vgcustom    1  1  0 wz--n- 1.98g  1.20g
vgdata      2  2  0 wz--n- 7.99g <3.00g

05:36:31 | root@localhost | ~ | cmd:198 | cpu: 0.35 | mem:28% | disk:21% | ip: | git: | ssh: | time:0s
🚀 lvs
LV          VG          Attr      LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
root        cs          -wi-ao---- <20.50g
swap        cs          -wi-ao---- 2.50g
lvnext      vgcustom    -wi-ao---- 800.00m
lvnext      vgdata      -wi-ao---- <3.00g
lvsize      vgdata      -wi-ao---- 2.00g

05:36:34 | root@localhost | ~ | cmd:199 | cpu: 0.32 | mem:28% | disk:21% | ip: | git: | ssh: | time:0s
🚀 pvs
PV          VG          Fmt  Attr PSize  PFree
/dev/sda3   cs          lvm2 a--  <23.00g  0
/dev/sdb1   vgdata      lvm2 a--  <4.00g  <2.00g
/dev/sdb2   vgdata      lvm2 a--  <4.00g  1.00g
/dev/sdb3   vgcustom    lvm2 a--  1.98g  1.20g

05:36:50 | root@localhost | ~ | cmd:200 | cpu: 0.55 | mem:28% | disk:21% | ip: | git: | ssh: | time:0s
🚀 df -h
Filesystem                Size  Used Avail Use% Mounted on
/dev/mapper/cs-root        21G   4.3G   17G   21% /
devtmpfs                   2.8G     0   2.8G    0% /dev
tmpfs                      2.8G   84K   2.8G    1% /dev/shm
tmpfs                      1.2G   9.5M   1.2G    1% /run
tmpfs                      1.0M     0   1.0M    0% /run/credentials/systemd-journald.service
/dev/sda2                  2.0G  446M   1.6G   23% /boot
tmpfs                      570M  116K  570M    1% /run/user/0
/dev/mapper/vgdata-lvsize  2.0G   71M   1.9G    4% /mnt/data1
/dev/mapper/vgdata-lvnext  3.0G   90M   2.9G    3% /mnt/data2
/dev/mapper/vgcustom-lvnext 736M   47M  690M    7% /mnt/backup
```

## LVM Management (Resize Operations)

### 5. Extend lvsize by 1GB

```
05:36:56 | root@localhost | ~ | cmd:201 | cpu: 0.51 | mem:28% | disk:21% | ip: | git: | ssh: | time:0s
🚀 lvextend -L +1G /dev/vgdata/lvsize
Size of logical volume vgdata/lvsize changed from 2.00 GiB (512 extents) to 3.00 GiB (768 extents).
Logical volume vgdata/lvsize successfully resized.

05:39:45 | root@localhost | ~ | cmd:202 | cpu: 0.14 | mem:28% | disk:21% | ip: | git: | ssh: | time:0s
🚀 xfs_growfs /mnt/data1
meta-data=/dev/mapper/vgdata-lvsize isize=512    agcount=4, agsize=131072 blks
       =                               sectsz=512   attr=2, projid32bit=1
       =                               crc=1        finobt=1, sparse=1, rmapbt=1
       =                               reflink=1     bigtime=1 inobtcount=1 nnext64=1
       =                               exchange=0    metadir=0
data    =                               bsize=4096   blocks=524288, imaxpct=25
       =                               sunit=0      swidth=0 blks
naming  =version 2                       bsize=4096   ascii-ci=0, ftype=1, parent=0
log      =internal log                   bsize=4096   blocks=16384, version=2
       =                               sectsz=512   sunit=0 blks, lazy-count=1
realtime =none                          extsz=4096   blocks=0, rtextents=0
       =                               rgcount=0    rgsz=0 extents
       =                               zoned=0      start=0 reserved=0
data blocks changed from 524288 to 786432
```

### 6. Resize the filesystem accordingly

### 7. Reduce lvext by 500MB (after unmounting)

### 8. Resize the filesystem safely before reduction

```
05:41:28 | root@localhost | ~ | cmd:203 | cpu: 0.15 | mem:28% | disk:21% | ip: | git: | ssh: | time:0s
🚀 umount /mnt/data2
```

```
05:44:10 | root@localhost | ~ | cmd:204 | cpu: 0.10 | mem:28% | disk:21% | ip: | git: | ssh: | time:0s
🚀 lvreduce -L 1.5G /dev/vgdata/lvnext
File system xfs found on vgdata/lvnext.
File system size (<3.00 GiB) is larger than the requested size (1.50 GiB).
File system reduce is required and not supported (xfs).
```

```
05:44:53 | root@localhost | ~ | cmd:205 | cpu: 0.04 | mem:28% | disk:21% | ip: | git: | ssh: | time:0s
🚀 mkfs.xfs -f /dev/vgdata/lvnext
meta-data=/dev/vgdata/lvnext  isize=512    agcount=4, agsize=196352 blks
       =                               sectsz=512   attr=2, projid32bit=1
       =                               crc=1        finobt=1, sparse=1, rmapbt=1
       =                               reflink=1     bigtime=1 inobtcount=1 nnext64=1
       =                               exchange=0    metadir=0
data    =                               bsize=4096   blocks=785408, imaxpct=25
       =                               sunit=0      swidth=0 blks
naming  =version 2                       bsize=4096   ascii-ci=0, ftype=1, parent=0
log      =internal log                   bsize=4096   blocks=16384, version=2
       =                               sectsz=512   sunit=0 blks, lazy-count=1
realtime =none                          extsz=4096   blocks=0, rtextents=0
       =                               rgcount=0    rgsz=0 extents
       =                               zoned=0      start=0 reserved=0
```

```
05:48:52 | root@localhost | ~ | cmd:209 | cpu: 0.1 | mem:28% | disk:21% | ip: | git: | ssh: | time:0s
🚀 mount /dev/vgdata/lvnext /mnt/data2
```

## Cleanup (WITHOUT reboot)

### 8. Unmount all logical volumes

```
05:45:55 | root@localhost | ~ | cmd:203 |  
:0s  
🚀 umount /mnt/data1  
  
05:48:24 | root@localhost | ~ | cmd:204 |  
:1s  
🚀 umount /mnt/data2  
  
05:49:26 | root@localhost | ~ | cmd:205 |  
:0s  
🚀 umount /mnt/backup
```

### 9. Remove entries from /etc/fstab

```
#  
# /etc/fstab  
# Created by anaconda on Sun Mar 15 19:41:01 2026  
#  
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.  
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.  
#  
# After editing this file, run 'systemctl daemon-reload' to update systemd  
# units generated from this file.  
#  
UUID=e79aa976-2f04-4e51-8942-08170334e649 / xfs defaults 0 0  
UUID=b41c8b19-dfe8-4088-8806-ec8fa03b629a /boot xfs defaults 0 0  
UUID=de55cc71-342c-4646-817d-ae6a99e96439 none swap defaults 0 0  
  
#/dev/sda1 /mnt/sda1 xfs defaults 0 0  
#-----  
  
~  
~  
~  
~
```

### 10. Remove Logical Volumes

```
05:52:04 | root@localhost | ~ | cmd:213 | cpu: 0.06 | mem:28% | disk:21% | ip: | git: | ssh: | t  
:0s  
🚀 lvremove /dev/vgdata/lvsize  
Do you really want to remove active logical volume vgdata/lvsize? [y/n]: y  
Logical volume "lvsize" successfully removed.  
  
05:52:45 | root@localhost | ~ | cmd:214 | cpu: 0.20 | mem:28% | disk:21% | ip: | git: | ssh: | t  
:0s  
🚀 lvremove /dev/vgdata/lvnext -y  
Logical volume "lvnext" successfully removed.  
  
05:52:54 | root@localhost | ~ | cmd:215 | cpu: 0.17 | mem:27% | disk:21% | ip: | git: | ssh: | t  
:1s  
🚀 lvremove /dev/vgcustom/lvnext -y  
Logical volume "lvnext" successfully removed.
```

## 11. Remove Volume Groups

```
05:53:08 | root@localhost | ~ | cmd:216 | cpu: 0.13 | mem:27% | disk:21% | t
:0s
🚀 vgremove vgdata
Volume group "vgdata" successfully removed

05:53:26 | root@localhost | ~ | cmd:217 | cpu: 0.10 | mem:28% | disk:21% | t
:0s
🚀 vgremove vgcustom
Volume group "vgcustom" successfully removed
```

## 12. Remove Physical Volumes

```
05:53:30 | root@localhost | ~ | cmd:218 | cpu: 0.09 | mem:28% | disk:21% | ip: | git: | ss
:0s
🚀 pvremove /dev/sd
/dev/sda3 /dev/sdb1 /dev/sdb2 /dev/sdb3

05:53:30 | root@localhost | ~ | cmd:218 | cpu: 0.09 | mem:28% | disk:21% | ip: | git: | ss
:0s
🚀 pvremove /dev/sd
/dev/sda3 /dev/sdb1 /dev/sdb2 /dev/sdb3

05:53:30 | root@localhost | ~ | cmd:218 | cpu: 0.09 | mem:28% | disk:21% | ip: | git: | ss
:0s
🚀 pvremove /dev/sdb*
Cannot use /dev/sdb: device is not in devices file
Labels on physical volume "/dev/sdb1" successfully wiped.
Labels on physical volume "/dev/sdb2" successfully wiped.
Labels on physical volume "/dev/sdb3" successfully wiped.
```

### 13. Delete partitions

```
Device      Boot      Start        End Sectors  Size Id Type
/dev/sdb1                2048    8390655  8388608    4G 8e Linux LVM
/dev/sdb2           8390656  16779263  8388608    4G 8e Linux LVM
/dev/sdb3           16779264  20971519  4192256    2G 8e Linux LVM

Command (m for help): d
Partition number (1-3, default 3): 1

Partition 1 has been deleted.

Command (m for help): d
Partition number (2,3, default 3):

Partition 3 has been deleted.

Command (m for help): d
Selected partition 2
Partition 2 has been deleted.

Command (m for help):

Command (m for help): p

Disk /dev/sdb: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x17c80010

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
```

### 14. Apply changes

```
05:56:27 | root@localhost | ~ | cmd:220 | cpu: 0
:0s
🚀 partprobe
```